

SYED MUHAMMAD MUJTABA ABBAS

Computer Science Undergraduate

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PROFESSIONAL SUMMARY

Computer Science undergraduate at PAF-KIET with hands-on experience gained through academic projects and a frontend development internship. Strong foundation in software engineering, data structures and algorithms, databases, computer networks, and information security. Experienced in developing practical software solutions using multiple programming languages and technologies, with a commitment to continuous learning and professional growth.

TECHNICAL SKILLS

Languages: C#, Java, Python, C, PHP, JavaScript

Frameworks & Libraries: React.js

Web Development: HTML, CSS, JavaScript, WordPress

Mobile Application Development: Java, Android Studio, XML, SharedPreferences

Databases: SQL, MySQL, Microsoft SQL Server, Firebase, Relational Database Design

Core CS Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), Software Engineering (SDLC), Database Management Systems (DBMS & ADBMS), Computer Networks, Information Security, Human-Computer Interaction (HCI), Artificial Intelligence (AI)

EXPERIENCE

Website Frontend Developer Intern

KDYS Lab — 3 Months

- Developed and maintained website frontend features using React.js
- Worked with PHP to support backend integration for web pages

PROJECTS

Career AI — AI-Powered Career Guidance Platform

Technologies: React.js, Python, FastAPI, Supabase, PostgreSQL, REST APIs, AI/LLM, JavaScript, HTML, CSS

- Developed a full-stack AI-powered career guidance platform using React.js for the frontend and Python/FastAPI for the backend.
- Integrated AI/LLM technology to provide personalized career guidance, skill-based recommendations, and interactive career assistance.
- Integrated Supabase as the database for securely storing and managing user and application data.
- Implemented user authentication and REST API communication between the frontend and backend.
- Built a responsive, modern, and user-friendly interface with properly integrated frontend, backend, AI services, authentication, and database functionality.

AI-Powered Chatbot Web Application

Technologies: React.js, Vite, Python, FastAPI, REST API, AI/LLM API, HTML, CSS, JavaScript

- Developed a modern AI-powered conversational web application with a responsive SaaS-style user interface.
- Built a React.js + Vite frontend for real-time user interaction, chat management, message rendering, and responsive UI/UX.
- Developed a Python FastAPI backend to handle API requests, process user messages, and communicate securely with an AI language model.
- Implemented RESTful API communication between the frontend and backend for seamless message exchange.
- Integrated an AI/LLM API to generate intelligent, context-aware responses to user queries.
- Implemented environment-based configuration for securely managing API credentials and backend settings.
- Designed the application using a frontend-backend architecture, separating presentation, API handling, and AI processing responsibilities.
- Added error handling for API failures, unavailable models, missing API credentials, and rate-limit/quota issues.
- Focused on creating a clean, professional, and user-friendly chatbot experience suitable for a modern AI SaaS product.
- Demonstrated practical experience in AI integration, backend API development, asynchronous request handling, React development, and full-stack application architecture.

AI Agent Platform — Intelligent Multi-Task AI Assistant

Technologies: React.js, Vite, Python, FastAPI, REST APIs, AI/LLM Integration, JavaScript, HTML, CSS

- Developed an AI Agent Platform designed to handle multiple types of user requests through a centralized intelligent interface.
- Built a modern React.js + Vite frontend with a responsive and interactive UI focused on usability and user experience.
- Developed a Python/FastAPI backend responsible for processing requests, executing application logic, and communicating with external APIs and AI services.
- Integrated AI capabilities to understand natural-language user requests and provide relevant responses.
- Designed the platform to handle real-world information queries, such as financial/stock information, sports schedules, calculations, and general knowledge requests.
- Implemented functionality for performing mathematical calculations and processing user commands through backend logic.
- Designed support for interacting with local desktop applications, such as launching applications like Spotify and Figma through system-level functionality.
- Structured the project around a modular architecture so additional AI capabilities and services can be integrated in the future.
- Implemented API-based communication between the frontend, backend, AI services, and external data sources.
- Applied environment variables and backend configuration practices for managing API keys and sensitive configuration securely.
- Focused on developing an extensible AI platform rather than a simple chatbot, combining AI processing, APIs, backend automation, and application-level functionality.
- Demonstrated practical knowledge of AI application development, API integration, backend engineering, automation, and modern web technologies.

Car Care Android Application

Android Studio · Java · XML · SharedPreferences

- Developed an Android application to help users manage basic car-related information, fuel records, maintenance, and expenses.
- Added a My Car section to save and load car details such as car name, model, year, and other information.
- Implemented a Fuel Tracker to record fuel quantity and price per litre, with automatic fuel cost calculation and total spending.
- Added Car Service and Maintenance functionality for recording services such as oil changes, brake service, and car washes.
- Developed an Expense Tracker to record expenses and calculate total spending.
- Created a Dashboard to display car information, total fuel cost, total expenses, and recent activities.
- Used SharedPreferences for storing and retrieving basic application data and XML layouts for a simple, user-friendly interface and navigation.

Hostel Management System — Web-Based Management System

Advanced Database Management Systems Project · PHP, MySQL

- Developed a web-based hostel management system using a 3-Tier Architecture with PHP and MySQL for better application organization and maintainability.
- Implemented parameterized queries and secure validation to prevent SQL injection in the login system, enhancing application security.

Shopping Management System — Desktop Application

Database Management Systems Project · C# Windows Forms, Microsoft SQL Server

- Designed and implemented a desktop-based shopping management system integrated with Microsoft SQL Server for efficient data storage and management.

Hospital Management System — Patient & Appointment Records

Data Structures & Algorithms Project · Python

- Developed a hospital management system using Python and core data structures to efficiently manage patient records, appointments, and hospital operations.

Clothing Brand Network Design & Simulation

Computer Networks Project · Cisco Packet Tracer

- Designed and simulated an enterprise network for a clothing brand, configuring routers, switches, VLANs, static routing, DHCP, DNS, and FTP for inter-network connectivity.

Language Model — Basic NLP Exploration

Artificial Intelligence Project · Python

- Developed a basic language model in Python to explore artificial intelligence concepts and natural language processing techniques.

Grocery Management System

Programming Fundamentals & OOP Projects · C#

- Built a console-based grocery management system for inventory management and billing, later redesigned using OOP principles (encapsulation, inheritance, polymorphism, abstraction) for a modular, maintainable application.

EDUCATION

Bachelor of Science in Computer Science (BSCS)

PAF-KIET (Karachi Institute of Economics and Technology), Karachi — 2023 – 2027

Intermediate (Pre-Engineering)

DHACCS Degree College, Karachi

SOFT SKILLS

Strong Problem-Solving, Fast Learner, Team Collaboration, Good Communication, Attention to Detail